

DS Written Homework 4

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1 AA Tree

2 Reb-Black Tree

3 The Height of Red-Black Tree

(a) Since red-black tree is a binary tree, the maximum number of nodes in a red-black tree of height h is $M_h = 2^{h-1} - 1$.

(b)

(c) Consider the node v in the red-black tree, let $h(v)$, $bh(v)$ be its height, black height, and $N(v)$ is the minimum nodes of the subtree with its root v .

Claim: the subtree with its root v has at least $N(v) = 2^{bh(v)-1} - 1$ nodes.

1. If $h(v) = 1$, v is NIL, we have $bh(v) = 1 \Rightarrow N(v) = 2^{1-1} - 1 = 0$
2. Induction assumption: $h(v) = k$ implies $N(v) = 2^{bh(v)-1} - 1$.
3. When $h(v') = k + 1$, the node v' has childs v_1, v_2 with there height is either $bh(v')$ or $bh(v') - 1$. By induction assumption we know

$$N(v_1) = N(v_2) = 2^{bh(v')-2} - 1$$

Hence, $N(v') = N(v_1) + N(v_2) + 1$ gives us

$$N(v') = 2^{bh(v')-2} - 1 + 2^{bh(v')-2} - 1 + 1 = 2^{bh(v')-1} - 1$$

By induction, the claim is proved.

Now suppose v is the root of the red-black tree. By the property of red-black tree, we have

$$\frac{h(v) - 1}{2} \leq bh(v) - 1 \leq h(v) - 1$$

The minimum nodes number occured when $bh(v) - 1 = \frac{h(v)-1}{2}$, which means

$$N_h = 2^{\frac{h-1}{2}} - 1$$

(d) With tree height h , we have $2^{\frac{h-1}{2}} - 1 \leq n \leq 2^{h-1} - 1$, which gives us

$$\log_2(n + 1) + 1 \leq h \leq 2 \log_2(n + 1) + 1$$

(e) Red-black tree height interval is $[\log_2(n+1) + 1, 2\log_2(n+1) + 1]$.

AVL tree height interval is $[\log_2(n+1) + 1, \log_\phi \sqrt{5}(n+1) - 2]$, where $\phi = \frac{\sqrt{5}+1}{2}$.

One can find that the RB tree interval is greater than AVL tree when n is large.

Hence, AVL trees enforces a stricter constraint on height.

4 2-4 Tree

5 Possible Red-Black Tree

(a)

(b)

(c)

6 Hash

(a)

(b)

(c)

(d)

7 Fill-in-the-blank questions

1.

2.

3.

4.

5.